

AARUSH JAGANNATHAN

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linkedin.com/in/aarush-jagannathan | github.com/ItzGOLLY | itzgolly.github.io/Portfolio

Final-year Electronics and Computer Engineering student with industry internships at Daimler Truck and Zoho Corporation. Experience spans manufacturing digitalisation and data analytics (Power BI, SQL, Advanced Excel), embedded IoT firmware (ESP32, Modbus RTU, MQTT over TLS) and backend systems (TypeScript, Express, PostgreSQL, REST APIs).

EDUCATION

B.Tech, Electronics and Computer Engineering | Vellore Institute of Technology (VIT), Chennai | 2023 – 2027

CGPA: 7.91 / 10 ($\approx$ 79.1%)

Sri Chaitanya Techno School | Class XII: 89% | Class X: 90%

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, TypeScript

Data & Analytics: Power BI, Advanced Excel (pivot models, weighted scoring matrices), data cleaning and structuring, dashboard design, KPI tracking

Backend & Databases: REST API design, Node.js / Express, PostgreSQL, JWT authentication, role-based access control, schema design, indexing, integration testing

Embedded & IoT: ESP32 / ESP32-S3, Modbus RTU over RS485, MQTT over TLS, UART / Serial, sensor integration, industrial energy metering, Arduino IDE

Tools & Methods: Git & GitHub, Linux, Postman, Cadence Virtuoso / Spectre, process mapping, requirement gathering, technical documentation

EXPERIENCE

Digitalisation Intern — Supplier Quality Development | Daimler India Commercial Vehicles (Daimler Truck AG), Oragadam | June – July 2026

- Tracked a 10-stage real-time data implementation rollout across 107 suppliers and built the Power BI dashboard the Supplier Quality Development team used to run and report it.
- Designed a Digital Transformation Priority Checklist scoring suppliers on 5 criteria mapped to 6 priority tiers, weighting cost and ROI highest based on research into Indian MSME capital constraints.
- Authored a Supplier Digitalisation Handbook and presented it at the Qprime supplier development conclave to an external supplier audience.
- Converted handwritten supplier submissions into structured datasets; delivered a 5-tab Excel priority-matrix workbook and a supplier-facing HTML scoring tool.

IoT Engineering Intern | Zoho Corporation, Chennai | May – June 2025

- Built an ESP32-based industrial IoT gateway that reads a Schneider EM6400NG three-phase energy meter over Modbus RTU on RS485 (9600 baud, 8N1) and publishes 9 electrical parameters every 5 seconds to Zoho IoT Cloud over MQTT with TLS.
- Decoded the meter's Modbus holding-register map to extract voltage, current, active/reactive/apparent power, power factor and frequency; configured 32 device data points cloud-side.
- Implemented Wi-Fi provisioning via a captive-portal access point so a field installer can commission the device from a phone with no laptop and no reflash; credentials persist in ESP32 NVS.
- Secured the cloud connection with an embedded root CA certificate. Stack: C / C++ (Arduino IDE), ModbusMaster, WiFiManager, Zoho IoT SDK, Preferences / NVS.

PROJECTS

ServiceDesk AI — Multi-Channel Support Platform | TypeScript, Express, PostgreSQL, React | github.com/ItzGOLLY

- Backend exposing ~45 REST endpoints with JWT authentication (bcrypt, httpOnly refresh-token cookies), role-based plus record-ownership authorisation and schema-level request validation.
- PostgreSQL 16 data model with referential constraints, indexes and GIN full-text search; ~127 unit and integration tests run against a live PostgreSQL instance.
- WhatsApp intake via Twilio with HMAC signature verification and database-level webhook idempotency, preventing duplicate tickets on provider retries.
- AI-assisted ticket classification behind a provider abstraction with a deterministic rule-based fallback, so intake degrades gracefully when the model API is down.

Ultra-Low-Power 10-bit SAR ADC | VLSI Project-I, 180 nm CMOS — Cadence Virtuoso / Spectre | 2026

- Comparative switching-energy study of four capacitive-DAC schemes (conventional, monotonic, Vcm-based, merge-and-split) on one testbench, benchmarked in normalised CV^2_{ref} .
- Literature survey of 20+ IEEE/JSSC papers to select a fabricated 7.6 nW, 1 kS/s reference design; evaluating INL/DNL, ENOB and Walden Figure of Merit. Review-I cleared September 2026.

ADDITIONAL

Achievement: Authored and presented the Supplier Digitalisation Handbook at Daimler Truck's Qprime supplier conclave (2026).